



Mediation Analysis with Compositional Exposures

Jing Ma

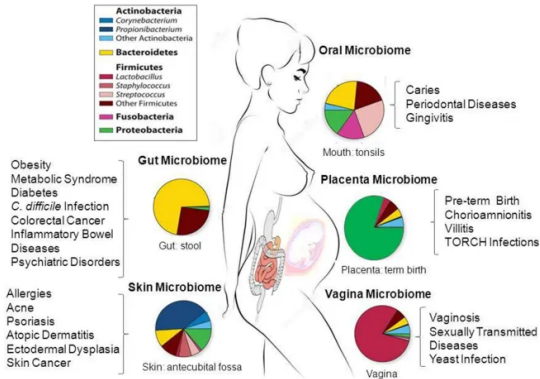
Public Health Sciences Division, Fred Hutch Cancer Center

Joint work with Poorbita Kundu, Timothy Randolph, Meredith Hullar and Sandi Navarro

August 3, 2026

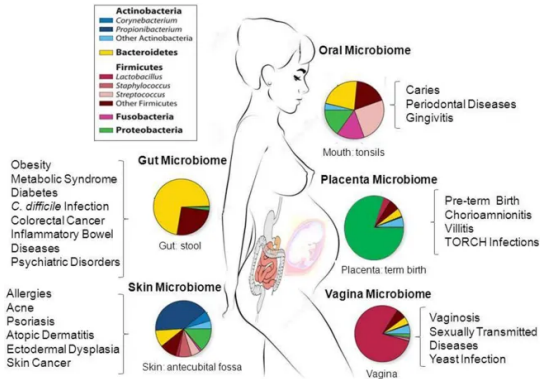
The human microbiome

Trillions of microbes carrying orders of magnitude more genes than the host genome, shaping metabolism, immunity, and inflammation (Belizário and Napolitano, 2015).



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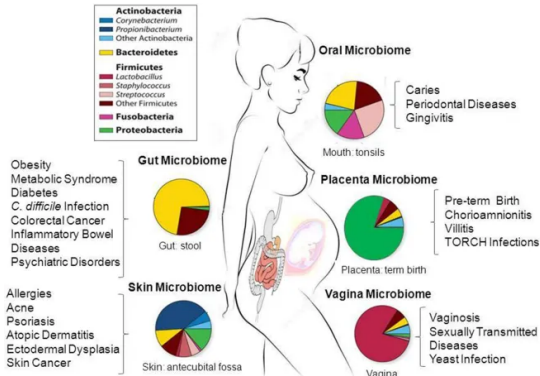
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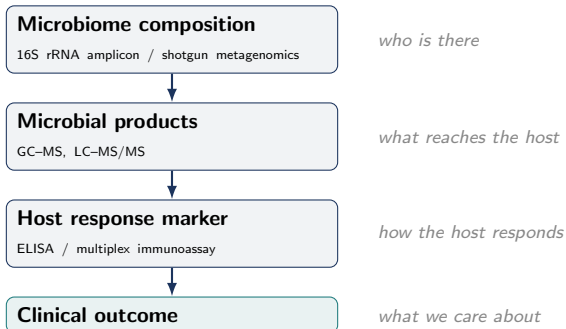
Mechanisms are what identify intervention targets.

Why multi-omics?

Composition tells us **who is there**; it does not tell us **how** the community acts on the host.

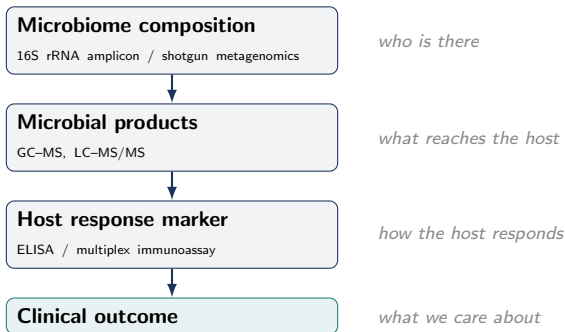
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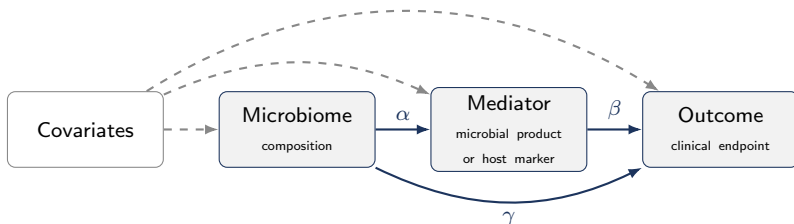
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Many cohorts, including the iHMP (Lloyd-Price et al., 2019) and MEC (Hullar et al., 2021), profile these layers on the *same* subjects — enough to ask a **mechanistic** question.

Mediation analysis

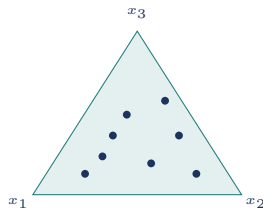
Does the effect of the microbiome on the outcome run **through** the intermediate layer?



We study the case where the microbiome is the **exposure** rather than the mediator.

Challenges when the microbiome is the *exposure*

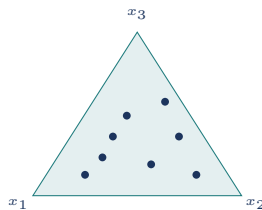
1. **Compositional.** Abundances carry only relative information (Gloor et al., 2017).



$$\sum_j x_j = 1$$

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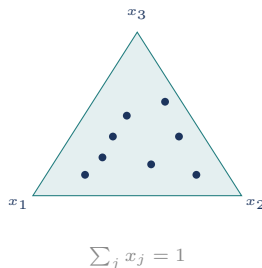
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2. **High-dimensional and sparse.** overdispersed counts, excess zeros.



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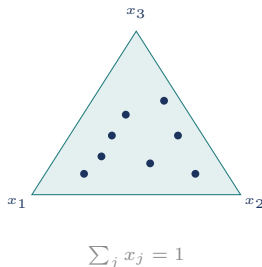
Challenges when the microbiome is the *exposure*

1. **Compositional.** Abundances carry only relative information (Gloor et al., 2017).
2. **High-dimensional and sparse.** overdispersed counts, excess zeros.
3. **No single interventional contrast.** A direct or indirect effect of a high-dimensional exposure is well defined only *after* the exposure is reduced to a low-dimensional summary (Kramer et al., 2026).



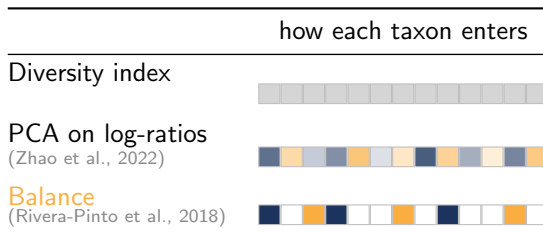
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How do we reduce dimension so that the latent exposure is
interpretable at the taxon level?

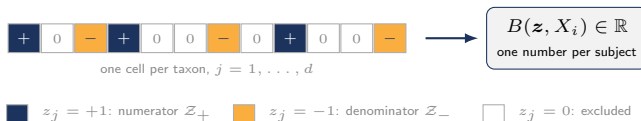
Dimensionality reduction



A balance is the **scalar** log-ratio between two geometric means

$$B(\mathbf{z}, X) = \sqrt{\frac{a_+ a_-}{a_+ + a_-}} \log \frac{g(X_{\mathcal{Z}_+})}{g(X_{\mathcal{Z}_-})}$$

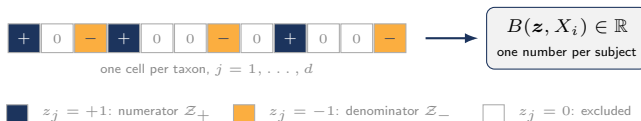
where $\mathbf{z} = (z_1, \dots, z_d)$ is an indicator vector such that $z_j \in \{+1, 0, -1\}$, $a_+ = |\mathcal{Z}_+|$ and $a_- = |\mathcal{Z}_-|$.



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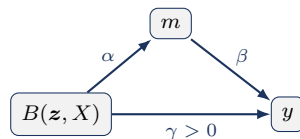


- Intervention can be done by modulating the balance (Rivera-Pinto et al., 2018).

Balance mediation model

Observed data $\{X_i, m_i, y_i\}_{i=1}^n$.

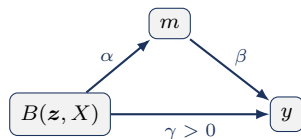
- $B(\mathbf{z}, X_i)$: latent scalar exposure
- m_i : scalar mediator (multivariate case later)
- y_i : continuous outcome



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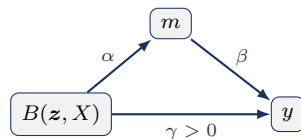
Linear SEM:

$$\begin{aligned}m_i &= \alpha B(\mathbf{z}, X_i) + e_i, & e_i &\sim N(0, \sigma_e^2), \\y_i &= \gamma B(\mathbf{z}, X_i) + \beta m_i + \varepsilon_i, & \varepsilon_i &\sim N(0, \sigma_\varepsilon^2).\end{aligned}$$

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$$y_i = \gamma B(\mathbf{z}, X_i) + \beta m_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma_\varepsilon^2).$$

Under the standard identification assumptions (VanderWeele, 2016), per unit change in the exposure:

$$\text{indirect (mediated) effect} = \alpha\beta, \quad \text{direct effect} = \gamma.$$

Unknowns: the balance z , effects (α, β, γ) , variances $(\sigma_e^2, \sigma_\varepsilon^2)$.

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Priors:

- independent 3-category prior on each taxon,

$$z_j \in \{+1, 0, -1\} \text{ w.p. } (\eta_1, 1 - \eta_1 - \eta_2, \eta_2);$$

smaller $\eta \Rightarrow$ sparser balance;

- conjugate normal-inverse-gamma on effects and variances;
- optional: LNM or LTN prior on X_i to capture sampling zeros.

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The balance is learned jointly with the effects.

Conditional on $\mathbf{z}$, everything is conjugate:

- $f(\alpha, \beta, \gamma \mid \mathbf{z}, \text{Data})$: normal
- $f(\sigma_e^2, \sigma_\varepsilon^2 \mid \mathbf{z}, \text{Data})$: inverse-gamma

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Collapsing improves mixing; the problem reduces to **sampling z** .

Update z one taxon at a time:

- sweep $j = 1, \dots, d$ in random order;
- draw $z_j \in \{+1, 0, -1\}$ from its 3-category full conditional (keeping $\mathcal{Z}_+, \mathcal{Z}_-$ non-empty);
- relabel the sign so that $\gamma > 0$.

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One sweep costs $O(d)$ density evaluations.

Metropolis–Hastings algorithm

Update value pairs (k, l) according to a local proposal $q_\tau(k, l)$ (Huang, 2020).

- Current state $\mathbf{z}$.
- Pick a pair of values $(k, l) \in \{(1, -1), (1, 0), (-1, 0)\}$ uniformly at random.
- Propose a new $\mathbf{z}'$ by modifying (k, l) according to q_τ .
- Compute the acceptance probability

$$\pi(\mathbf{z}', \mathbf{z}, k, l) = \min \left(1, \frac{f(\mathbf{z}') q_\tau(k, l \mid \mathbf{z}')}{f(\mathbf{z}) q_\tau(k, l \mid \mathbf{z})} \right), \quad k \neq l.$$

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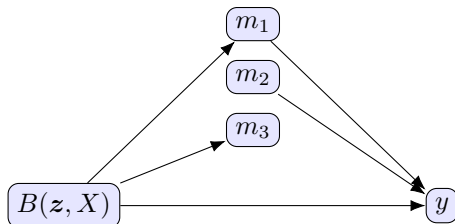
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Stochastic search is fast, but acceptance rate can be low for large d .

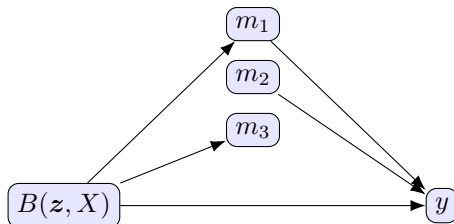
Multivariate mediators

Real pathways involve **several correlated** mediators.



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- mediator model with a **free residual covariance**;
- select active mediators *jointly* with the balance, adapting the **coordinated selection** of correlated mediators (Song et al., 2021).

Data generating model

- $d = 50$ taxa; zero-inflated log-normal counts ($\sim 40\%$ zeros), total-sum normalized.
- True balance $\mathbf{z} = (+1, +1, +1, -1, -1, -1, 0, \dots, 0)$.
- True effects $\alpha = \gamma = 1/3$, $\beta \in \{0, 2\} \Rightarrow \text{IE} = \alpha\beta \in \{0, 2/3\}$.

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- **Scenarios:**
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- **Competitors:**
 - **oracle** (true $\mathbf{z}$)
 - unsupervised principal balance
 - regDOC (Wang et al., 2019) on CLR-transformed data

Results: indirect effect (Scenario I)

BalMed tracks the oracle; an outcome-blind balance does not.

	Bias	RMSE	Cover ₉₅	Reject
<i>Null ($\beta = 0$)</i>				
BalMed	-0.04	0.06	0.73	0.27
Oracle	0.00	0.03	0.96	0.04
Principal balance	0.00	0.02	—	—
<i>IE = 2/3 ($\beta = 2$)</i>				
BalMed	0.13	0.21	0.59	0.97
Oracle	0.00	0.12	0.93	1.00
Principal balance	-0.67	0.69	—	—

Sample size $n = 100$. Gap to the oracle is calibration, and shrinks at $n = 200$.

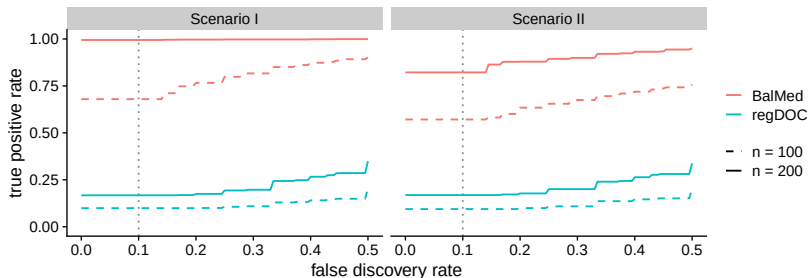
Results: indirect effect (Scenario II)

Same pattern under **misspecified balance weights** — BalMed is robust to the loading pattern.

	Bias	RMSE	Cover ₉₅	Reject
<i>Null ($\beta = 0$)</i>				
BalMed	-0.02	0.05	0.81	0.19
Oracle	0.02	0.04	0.93	0.08
Principal balance	0.00	0.02	—	—
<i>IE = 2/3 ($\beta = 2$)</i>				
BalMed	0.15	0.22	0.62	0.95
Oracle	-0.04	0.12	0.93	1.00
Principal balance	-0.67	0.69	—	—

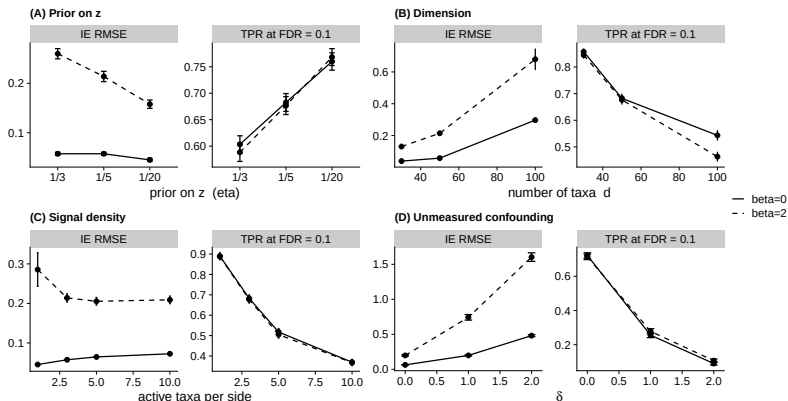
Under the null, even the oracle's Type I error is mildly inflated by the misspecified weights.

Results: taxon selection



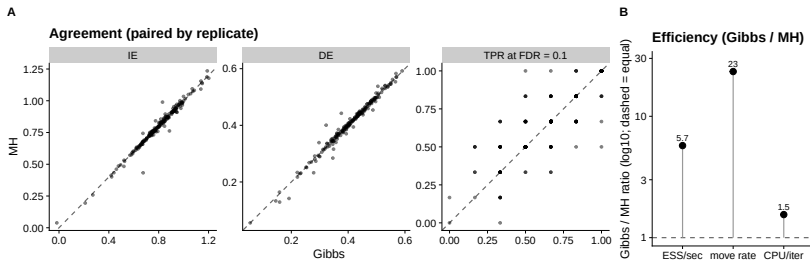
- BalMed **dominates** regDOC across the range.
- At FDR = 0.1: recovers **57–99%** of active taxa vs. **9–17%** for regDOC.

Sensitivity analysis



- Robust to the prior, dimension, and sparsity of z .
- Breaks down under **unmeasured mediator–outcome confounding**, as the identification theory predicts.

Gibbs vs. Metropolis–Hastings



- Same answers (points on $y = x$).
- Gibbs is far more efficient: $\approx 5.7\times$ the effective samples per second.

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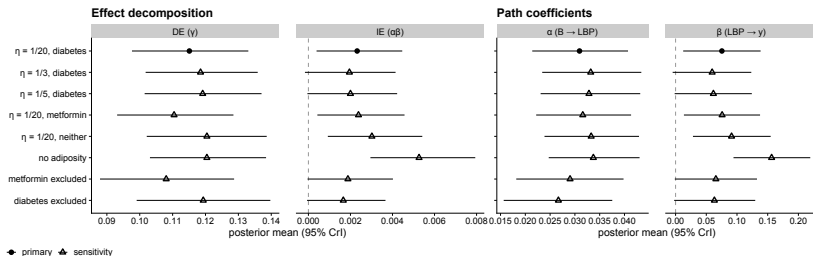
- Multiethnic Cohort–Adiposity Phenotype Study (Hullar et al., 2021); $n = 1513$ older adults, 5 ethnic groups.
- **Exposure:** gut microbiome (135 genera).
- **Mediator:** lipopolysaccharide-binding protein (marker of endotoxemia).
- **Outcome:** percent liver fat (MRI).

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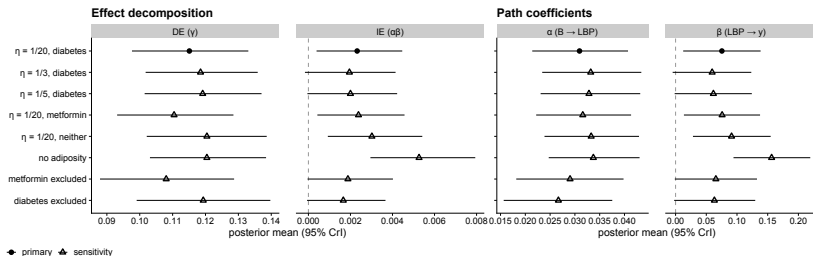
Adjusted for demographics, total body-fat percentage, diet, behavior, season, and diabetes status. Sensitivity fits vary the prior, swap diabetes for metformin use (or neither), drop the adiposity adjustment, or exclude metformin users or participants with diabetes.

MEC-APS: effect decomposition



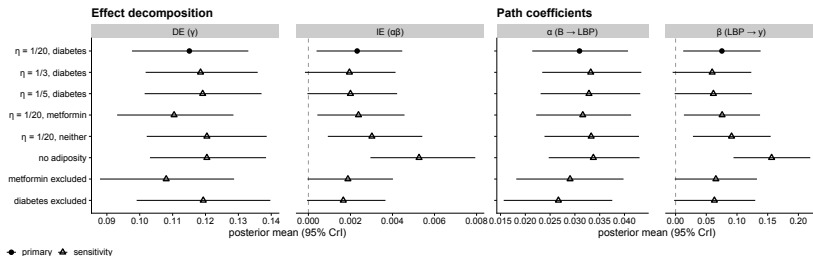
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- Small **positive** indirect effect through LBP, $\alpha\beta = 0.0023$ (0.0004 to 0.0045): about 2% of the total effect.
- The indirect effect's CrI includes zero under less sparse priors and in the exclusion fits; dropping the adiposity adjustment **roughly doubles** it.

MEC-APS: taxon selection

21 genera selected (PIP > 0.5) in the primary fit; 10 are selected in all eight fits.

Numerator (higher liver fat, LBP)		Denominator	
Lachnospiraceae (uncultured)	+1.00	Clostridia UCG-014	−1.00
<i>Marvinbryantia</i>	+0.99	<i>Oscillospira</i>	−1.00
<i>Lachnoclostridium</i>	+0.96	<i>Erysipelatoclostridium</i>	−0.97
<i>Veillonella</i>	+0.91	<i>Intestinimonas</i>	−0.95
<i>Bacteroides</i>	+0.89	<i>Haemophilus</i>	−0.90
Lachnospiraceae CAG-56	+0.76	Lachnospiraceae NK4A136	−0.89
<i>Howardella</i>	+0.74	Christensenellaceae (uncultured)	−0.82
<i>Colidextribacter</i>	+0.73	<i>Anaerotruncus</i>	−0.66
<i>Negativibacillus</i>	+0.67	<i>Holdemania</i>	−0.65
<i>Ca. Stoquefichus</i>	+0.60	<i>Cloacibacillus</i>	−0.51
<i>Bilophila</i>	+0.60		

- Denominator: mostly obligate anaerobes, several SCFA producers — in line with the butyrate-producer depletion reported in fatty liver disease (Aron-Wisniewsky et al., 2020).
- Numerator: Gram-negative *Bacteroides*, *Veillonella*, *Bilophila*, and several Lachnospiraceae.

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Caution: the direct effect is an order of magnitude larger, so the balance mainly reflects the direct association; data are observational.

- Bayesian mediation analysis when the exposure is a **composition**.

Summary

- Bayesian mediation analysis when the exposure is a **composition**.
- A **balance** reduces the composition to an interpretable scalar log-ratio — learned *jointly* with the effects.

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- A **balance** reduces the composition to an interpretable scalar log-ratio — learned *jointly* with the effects.
- Collapsed Gibbs/MH sampler.
- MEC-APS: a robust direct association with liver fat, and a small indirect effect through LBP that depends on adiposity adjustment.



Thank you!

Preprint: bioRxiv, doi:10.64898/2026.09.24.754147

Code: github.com/drjingma/BalExMed

This work was supported by NIGMS.

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